

IUSUP ABDULIAPAROV

Unity C# Engineer | Medical Simulation, Real-Time Rendering and Physics

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PROFESSIONAL SUMMARY

Unity C# engineer specialising in real-time simulation and visualization. Currently the sole engineer on a coronary angiography trainer, building its fluoroscopic imaging, contrast haemodynamics, instrument physics and the angioplasty and stenting workflow, validated against reference angiograms with clinical reviewers. Earlier work shipped high-traffic real-time products reaching 9M+ users at 40k DAU peak, cutting WebGL build size from 100 MB to 20 MB and lifting low-end Android from ~15 to a stable 60 FPS.

WORK EXPERIENCE

Coronary Angiography Simulator

Unity Simulation Engineer

2025 to Present

Sole engineer on a coronary angiography trainer used to rehearse catheter procedures: X-ray imaging, contrast behaviour, instrument physics and the balloon and stent workflow, matched to reference angiograms with clinical reviewers.

- Delivered the X-ray imaging pipeline as a five-pass URP render feature (per-tissue density accumulation, detector gain and lag, DSA subtraction road-mapping and a CT volume path), tuned frame by frame against reference angiograms.
- Made contrast behave like injected dye rather than an animation: advection-diffusion transport solved on a 19-segment vessel graph built from the artery mesh, so filling follows anatomy, slows through a stenosis and stops at an occlusion.
- Cut the injection frame hitch 2.6x, from 12.3 ms to 4.7 ms, through profiler-driven work on the graph solver, holding real-time rates while the physics solver already consumed 59% of the frame.
- Kept injection honest under motion: independent boluses superpose in a single field and the still-emitting volume tracks the catheter tip through a breathing, beating vessel.
- Built instrument physics on a tail-driven rod solver with lumen coupling so catheter and guidewire move coaxially, with 8 instrument profiles defined by configuration rather than code.
- Modelled treatment as it is performed: the balloon opens the lesion and elastically recoils, while a stent crimped on it expands, implants on deflation and stays fixed to the breathing vessel wall.

Tech: Unity, C#, URP, ShaderLab/HLSL, Burst, custom render features, ScriptableObjects, procedural meshes.

Aviana Studios, WebGL and Mobile Products

Unity Engineer | Engineering Owner and Co-founder

Remote

2023 to 2026

Owned Unity engineering for a high-traffic Unity WebGL product reaching 9M+ users at 40k DAU peak, plus nine standalone WebGL titles built on a shared codebase.

- Built and maintained a shared Unity codebase shipping nine standalone WebGL titles (Find the Brainrots, Plane Builder, Plus-One Jump, Plus-One Run, Slimes, Build Island and others) from per-game feature branches; reusable services, state machine and save layer across all of them.
- Reduced WebGL build size from 100 MB to 20 MB through shader-variant scoping, asset stripping and texture/audio re-encoding; improved low-end Android from ~15 FPS to stable 60 FPS via GC allocation reduction, pooling, batching and profiler-driven optimization; eliminated mobile-browser page reloads using on-demand Resources.Load with explicit UnloadUnusedAssets / GC.Collect between scene transitions.
- Built a 1k-line custom Rigidbody vehicle controller: spring-damper suspension with bump stops and sphere-cast wheel probes, drift, boost, downforce, air control and auto-recovery; refactored from an earlier ConfigurableJoint prototype to stabilize handling for the game's arcade feel.
- Designed a reusable Unity architecture: DI (VContainer / custom service locator), hierarchical state machines, ScriptableObject configuration and a multi-backend save abstraction (PlayerPrefs / Mirra SDK cloud sync) driving reactive UI through R3 ReactiveProperty.
- Integrated production monetization stack: AppLovin MAX with 13 mediation adapters, Unity IAP, Google Play Billing and Play Core in-app updates, plus five analytics SDKs (GameAnalytics, AppMetrica, AppsFlyer, ByteBrew, Firebase) behind a single facade.
- Optimized environment geometry from ~1M to ~40k triangles on a mobile prototype via mesh decimation, LODs and silhouette-preserving proxies; supported production releases across WebGL and mobile platforms under strict performance, memory and build-size constraints.

Tech: Unity, C#, URP, VContainer, R3, UniTask, Addressables, AppLovin MAX, Mirra SDK, Playgama.

TECHNICAL SKILLS

- **Engine & Languages:** C#, Unity (URP)
- **Architecture:** SOLID, dependency injection (VContainer, custom service locator), hierarchical state machines, ScriptableObject-driven configuration, reactive event systems (R3, UniRx, UniTask), save-system abstraction with multi-backend implementations
- **Rendering:** custom URP render features and passes, image-effect composition, X-ray-style visualization, WebGL shader budgeting, ShaderVariantCollection management, Cinemachine 3, Unity Animation Rigging

- **Physics & Simulation:** custom Rigidbody vehicle controller (spring-damper, sphere-cast, drift/boost/air-control), ConfigurableJoints, guidewire/catheter simulation, multi-rigidbody systems, procedural meshes, blendshape-driven deformation
- **Platform SDKs:** AppLovin MAX (13 mediation adapters), Unity IAP, Google Play Billing & Play Core, Mirra SDK 5, Playgama Bridge (WebGL portals)
- **Analytics:** GameAnalytics, AppMetrica, AppsFlyer, ByteBrew, Firebase Analytics
- **Performance:** Unity Profiler, Memory Profiler, GC allocation reduction, draw-call analysis and batching, WebGL build-size optimization, low-end Android profiling
- **Production Tooling:** Addressables, Unity Localization, custom Editor tools, content pipelines, pre-build validation, Git

EDUCATION

Università Cattolica del Sacro Cuore

BSc Economics & Management

Ufa State Petroleum Tech. University

Automation & Control Systems

Milan, Italy

2024 to Present

Ufa, Russia

2022 to 2024